RUTGERS

# Phonetics I

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2026/11/09 (updated: 2026-11-07)

Please pick up your **name card** from the front

**office hours**: Mondays, 2:00-3:00 pm

Basement, Room 003, **Linguistics Building**

# Previously on 201




## Pragmatics

- the study of how **context** contributes to meaning or it is the study of **implied meaning**

## presuppositions

- background information that is taken for granted or in the **common ground** for an utterance to make sense

## conversational implicatures

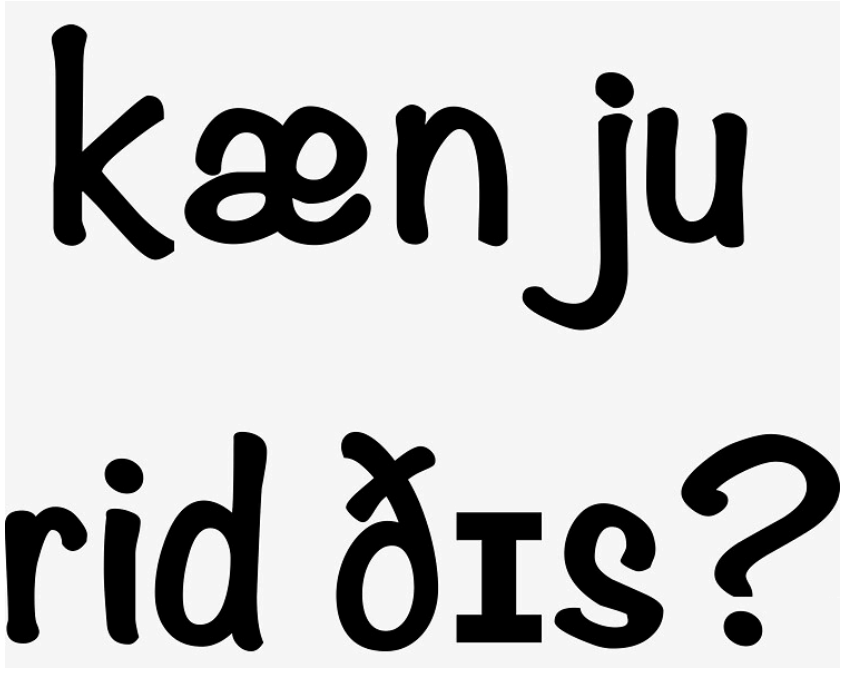
- meaning arises from our understanding of **how people behave** in conversation

## Gricean Maxims

- quality, **quantity**, relevance, **manner**


NEVER EVER
BE FRIGHTENED BY
PHONETICS
AT ALL

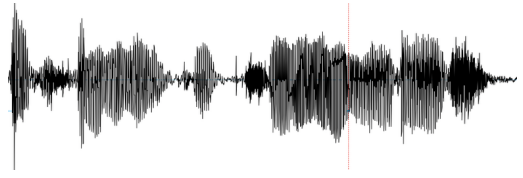
# Sounds of Language




- the **smallest** element of language: **sounds**
- language can be spoken, written, signed, etc
- the use of voice (=speech) remains the most common modality used for language
- human have an anatomic specialization for speech and production

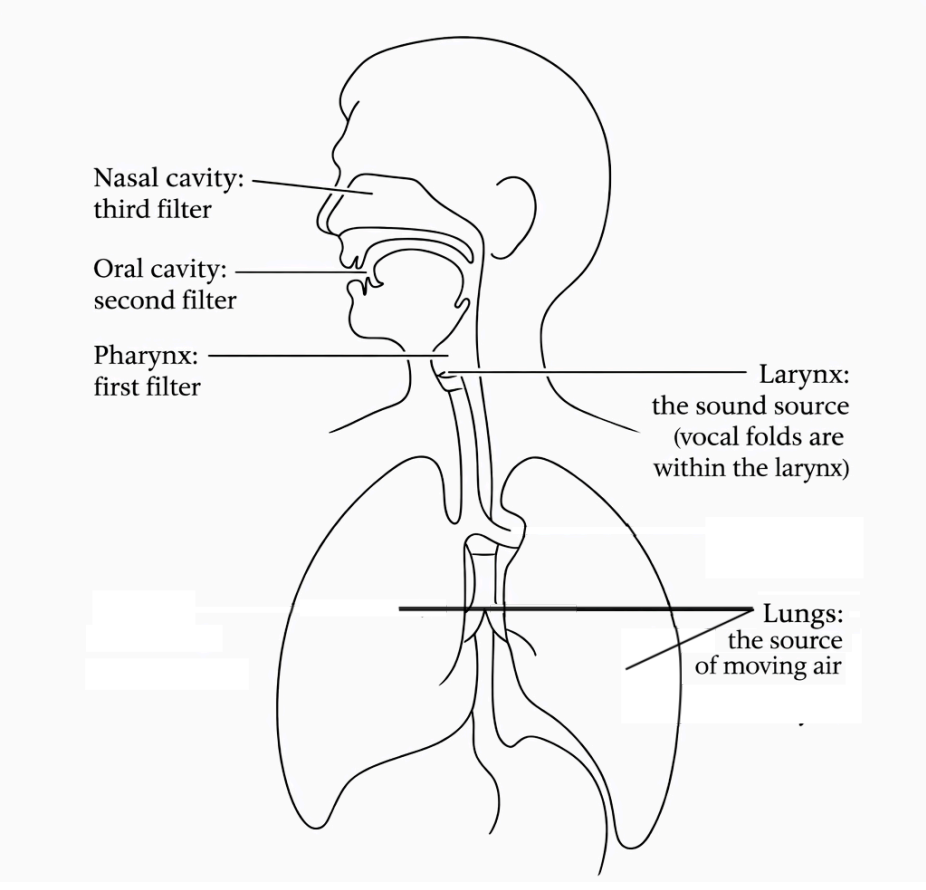
an MRI video of English production

an MRI video of Mandarin Chinese production



# Phonetics vs Phonology

- **Phonetics** is the study of **how speech sounds are produced** by articulators in the vocal tract and **perceived** by are auditory apparatus
    - **articulatory phonetics**: how different sounds are produced in vocal tract
    - **acoustic phonetics**: the physical properties of speech sounds
    - **perceptual phonetics**: how sounds are perceived by listeners
- **Phonology** is the study of the **pattern** and **function** of sounds (how your brain categorises sounds associated with speech)



# Articulation: How Do We Make Speech Sounds?

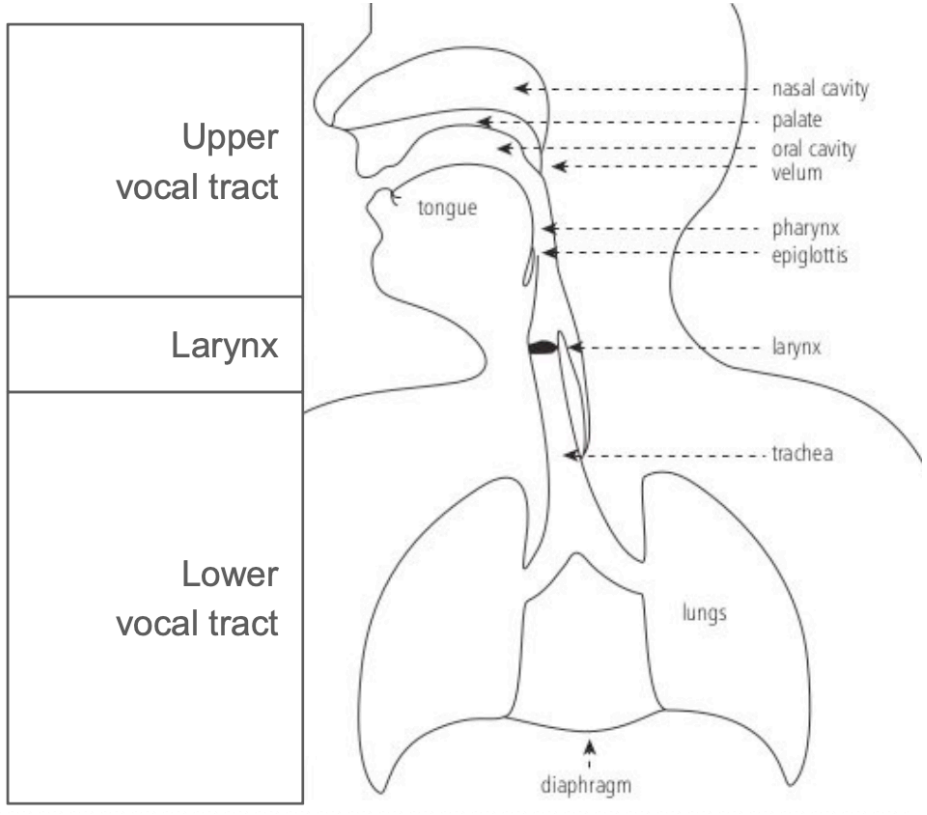
**articulation**: moving parts of body to produce speech sounds

1. speaker **exhales**, pushing air out of **lungs** and up the **trachea** (windpipe)

2. air passes through **larynx** (voicebox), which contains **vocal folds** (cords), primary source of speech sounds

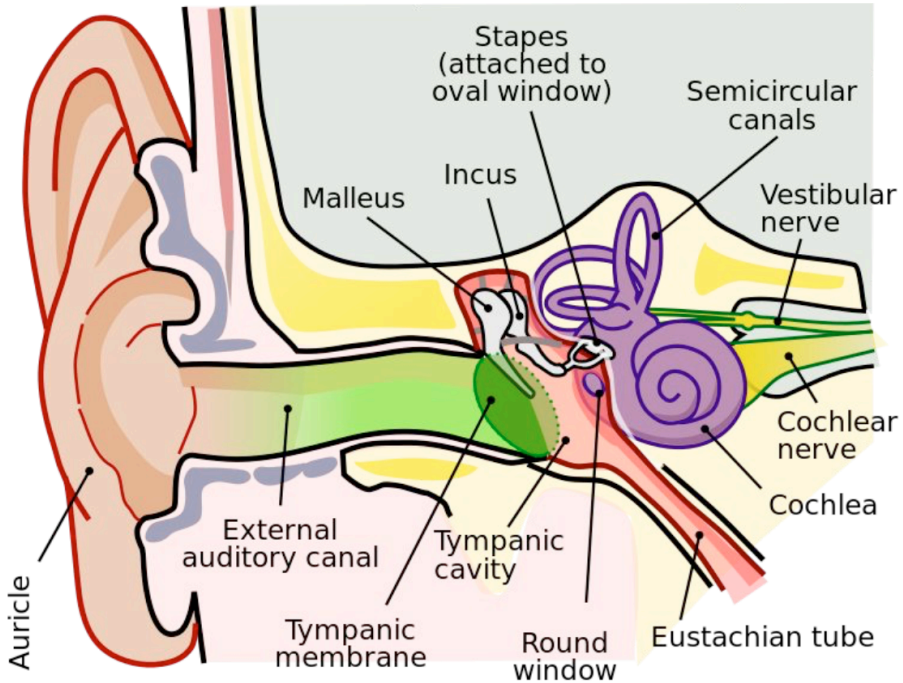
3. air passes through **upper vocal tract**, including **oral cavity** (mouth) and **nasal cavity** and leaves body

4. within the vocal tract, **articulators** such as the tongue, lips, teeth, and soft palate move to **modify** the sound coming from the larynx and produce additional sounds



# Perception: How Do We Hear Sounds?

perception involves both **ear** and **brain**




- sound waves are channeled by the outer ear through the ear canal to the **eardrum**, causing it to **vibrate**
- these vibrations are transmitted by the ossicles and tympanic cavity to the **inner ear**
- the inner ear is full of **fluid** and vibrations pass through the fluid into the spiral-shaped **cochlea**
- at the cochlea, the vibrations cause movements in a series of tiny **hairs** – different hairs for different pitches
- and the hairs connect to **nerves**, which send neural impulses to the brain, where sounds are decoded into language

# Segments and Supra-segmentals

- **segment**: discrete, **individual** unit of sound
  - for example a consonant, vowel, or glide
  - speech stream can be cleanly divided into segments: **< cats >** = [k], [æ],[t], [s]

- but some sounds occur **above the level of the segment**, affecting **multiple** segments
  - prosodic sounds or **suprasegmentals**, for example, **stress**

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**fantastic**

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**fantastic** **fanTAStic**

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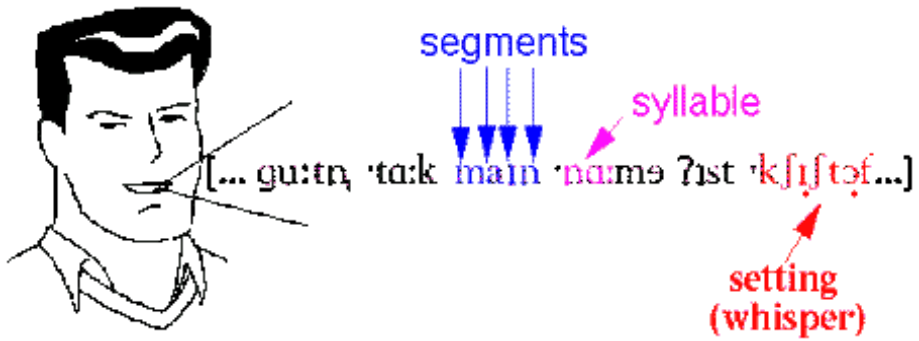
**fantastic** **fanTAStic**

- **fanTAStic** – stressed syllables tend to be louder, longer, more clearly enunciated, and pronounced with a change in pitch

- this affects **all** sounds in the syllable **[ tas ]**

- we'll focus on sound **segments** in this class

# Segments




say these words out loud and try breaking them down into **segments**

(1). what and how many segments does each word have?
(2). what letters do they correspond to and does one letter always correspond to one sound?

- banana
- ship
- peak
- knock
- rough

# Letters ≠ Sounds

- different languages also use different alphabets to represent sounds
- but alphabets are not good reflections of sounds, therefore we distinguish **orthography** from **pronunciation**

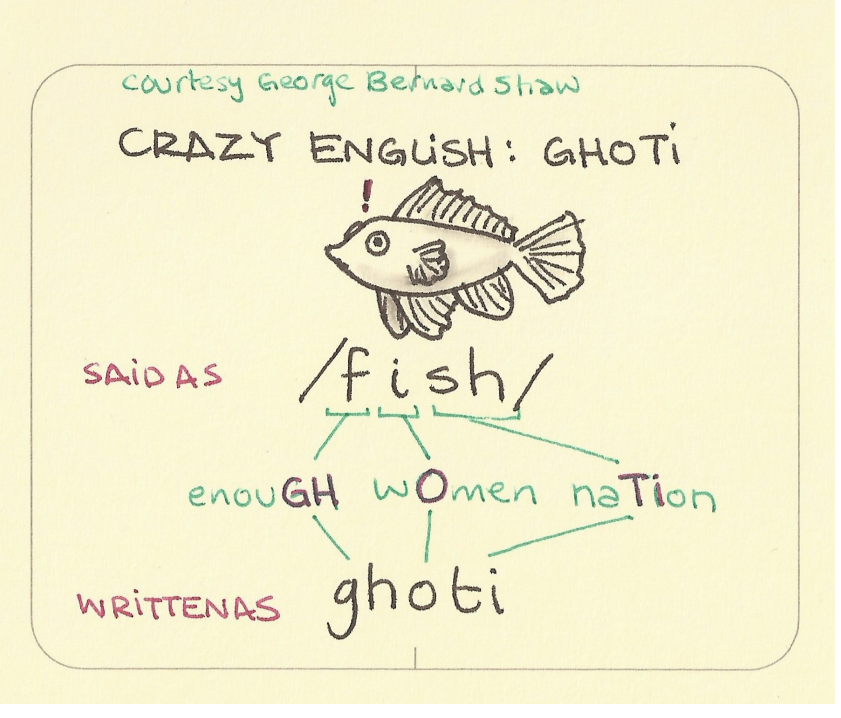
**< letter >** = orthography; **[ phonetic symbol ]** = pronunciation

| | words in different language | < letter > | [ phonetic symbol ] |
|---|---|---|---|
| 0:00 / 0:02 English | *r*egard | < r > | [ ɹ ] |
| 0:00 / 0:02 French | *r*egarder ('look') | < r > | [ ʁ ] |
| 0:00 / 0:02 Mandarin | *r*è gǒu ('hot dog') | < r > | [ ʐ ] |
| 0:00 / 0:02 Spanish | *r*egalo ('present') | < r > | [ r ] |

# Letters ≠ Sounds

**spelling** (orthography) has a problem: one sound doesn't always equal one letter

- two letters, one sound
  - ***th*** and ***ng*** in ***thing***
- one letter, two sounds
  - ***x*** in *ta**x***
- one letter, no sound
  - ***k***, ***g***, and ***h*** in ***k**nig**h**t***
- one sound, many spellings
  - *rush*, *notion*, *ocean*, *niche*, *issue*, *schlep*, *fuchsia*
- one spelling, many sounds
  - *thr**ough** ≠ pl**ough** ≠ **ought** ≠ r**ough** ≠ c**ough** ≠ th**ough***



# Letters ≠ Sounds




- no language has a writing system that can **perfectly** represent its speech sounds

- pronunciation **changes over time**; writing doesn't always reflect these changes
  - *knight*, *time*
- pronunciation **varies by place and social group**; writing doesn't usually reflect this
  - *schedule*, *cot/caught*
- **loanwords** (words borrowed from another language) often have spellings that reflect the pronunciation in the source language, not target language
  - *zucchini*, *psychology*

- this becomes a problem when we want to talk precisely about **phonetics**, and we want to accurately represent the **pronunciation**

# An Easy Cryptographic Game

A: I
B: II
C: III
D: IV
E: V
F: VI
G: VII
H: VIII
I: IX
J: X
K: XI
L: XII
M: XIII
N: XIV
O: XV
P: XVI
Q: XVII
R: XVIII
S: XIX
T: XX
U: XXI
V: XXII
W: XXIII
X: XXIV
Y: XXV
Z: XXVI

IX XVI I

XVIII XXI XII V XIX

XXI XIX

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IX XVI I

XVIII XXI XII V XIX

XXI XIX




*'IPA rules us.' (not U-S)*

# International Phonetic Alphabet (IPA)



# International Phonetic Alphabet (IPA)

## [ˌɪnəɹˈnæʃənəl fəˈnɛɪɪk ˈælfəˌbɛt]

- to **accurately** represent speech sounds, we use the **International Phonetic Alphabet (IPA)**
  - one symbol **=** one sound
- same symbols for **all** spoken languages – all languages use some **subset** of these symbols (plus additional symbols on the full IPA chart)
- many IPA symbols represent roughly the same sounds as the corresponding English letter
  - [p, t, k, b, d, g, f, v, s, z, h, m, n, l]
- some IPA symbols do not
  - IPA [**j**] = English y, *yes* [jɛs]
  - [**c**, **q**, **x**] & most vowels ≠ English letters

CONSONANTS (PULMONIC) ©℗® 2020 IPA

| | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Plosive | p b | | | t d | | ʈ ɖ | c ɟ | k g | q ɢ | | ʔ |
| Nasal | m | ɱ | | n | | ɳ | ɲ | ŋ | ɴ | | |
| Trill | ʙ | | | r | | | | | ʀ | | |
| Tap or Flap | | ⱱ | | ɾ | | ɽ | | | | | |
| Fricative | ɸ β | f v | θ ð | s z | ʃ ʒ | ʂ ʐ | ç ʝ | x ɣ | χ ʁ | ħ ʕ | h ɦ |
| Lateral fricative | | | | ɬ ɮ | | | | | | | |
| Approximant | | ʋ | | ɹ | | ɻ | j | ɰ | | | |
| Lateral approximant | | | | l | | ɭ | ʎ | ʟ | | | |

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

CONSONANTS (NON-PULMONIC)

| Clicks | Voiced implosives | Ejectives |
|---|---|---|
| ʘ Bilabial | ɓ Bilabial | ʼ Examples: |
| ǀ Dental | ɗ Dental/alveolar | pʼ Bilabial |
| ǃ (Post)alveolar | ʄ Palatal | tʼ Dental/alveolar |
| ǂ Palatoalveolar | ɠ Velar | kʼ Velar |
| ǁ Alveolar lateral | ʛ Uvular | sʼ Alveolar fricative |

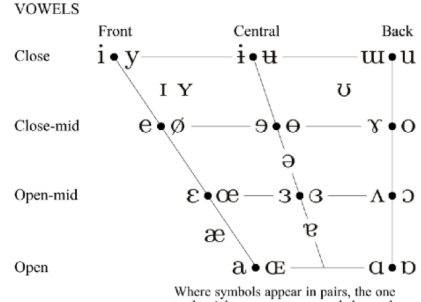
OTHER SYMBOLS

ʍ Voiceless labial-velar fricative
w Voiced labial-velar approximant
ɥ Voiced labial-palatal approximant
ʜ Voiceless epiglottal fricative
ʢ Voiced epiglottal fricative
ʡ Epiglottal plosive
ɕ ʑ Alveolo-palatal fricatives
ɺ Voiced alveolar lateral flap
ʧ Simultaneous ʃ and x

Affricates and double articulations can be represented by two symbols joined by a tie bar if necessary.

t͡s k͡p

VOWELS




Where symbols appear in pairs, the one to the right represents a rounded vowel.

# International Phonetic Alphabet (IPA)

[ˌɪnəɹˈnæʃənəl fəˈnɛɪɪk ˈælfəˌbɛt]

CONSONANTS (PULMONIC)

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| Tap or Flap | | | | ɾ | | | | |
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| Approximant | | | | ɹ | | j | | |
| Lateral approximant | | | | l | | | | |

ʍ Voiced labial-velar approximant

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DIACRITICS

| | | |
|---|---|---|
| ◌̥ | Voiceless | n̥ d̥ |
| ◌ʰ | Aspirated | tʰ dʰ |
| ◌̩ | Syllabic | n̩ |
| ◌˞ | Rhoticity | ə˞ a˞ |
| ◌̴ | Velarized or pharyngealized | ɫ |
| ◌̃ | Nasalized | ẽ |

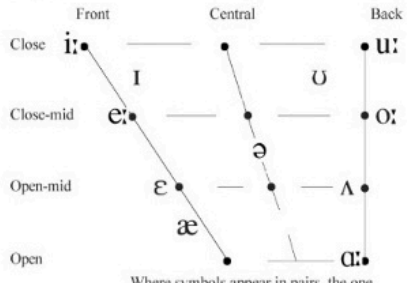
SUPRASEGMENTALS

| | |
|---|---|
| ˈ | Primary stress ˌfoʊnəˈtɪʃən |
| ˌ | Secondary stress |
| ː | Long eː |
| . | Syllable break ɹi.ækt |

OTHER SYMBOLS

Affricates and double articulations can be represented by two symbols joined by a tie bar if necessary. k͡p t͡s

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| Tap or Flap | | ⱱ | | ɾ | | ɽ | | | | | |
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| Lateral fricative | | | | ɬ ɮ | | | | | | | |
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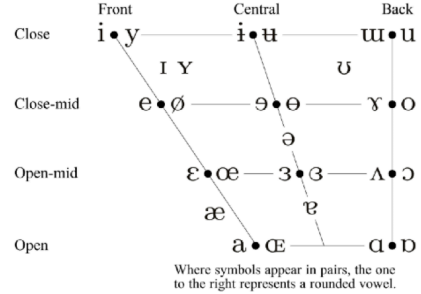
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| ʡ Epiglottal plosive | |

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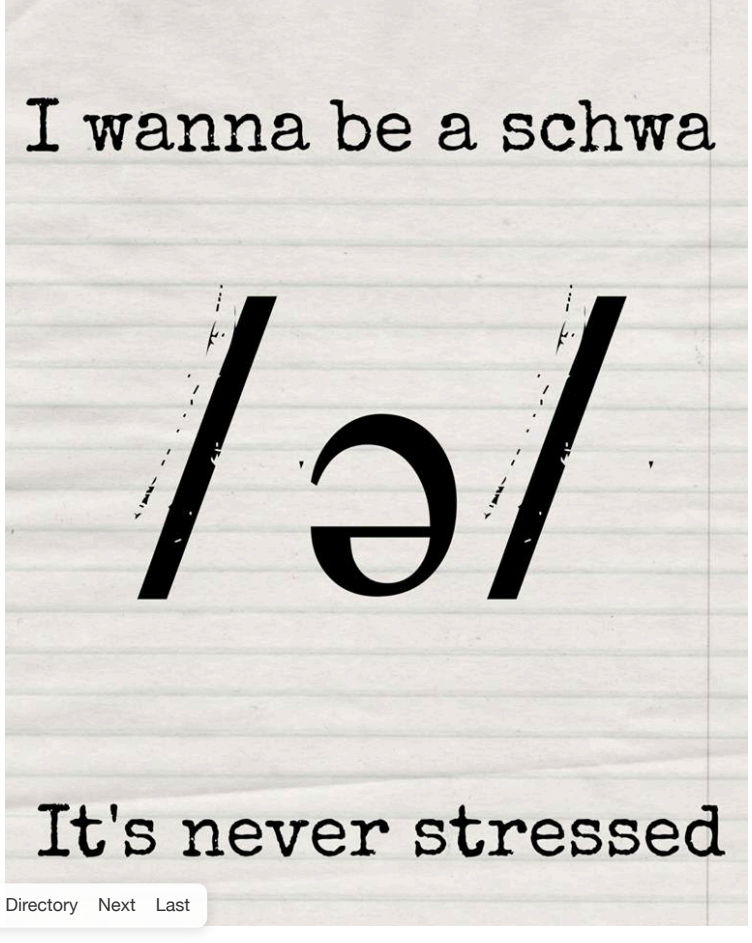
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# International Phonetic Alphabet (IPA)

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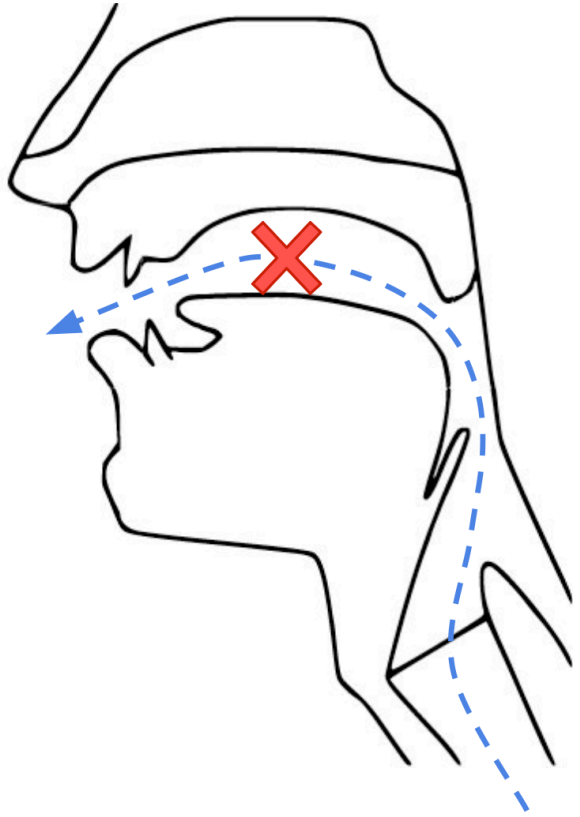



## conventions

1. phonetic transcriptions are put between **square brackets [**bə.ˈnæ.nə**]**
2. spelling (orthography) is written in ***italics*** or **angle brackets**: *banana*, **<b>**
3. **all** sounds and **only** sounds are written:
   - no silent letters
     **ph**on**e** [ˈfown]; throu**gh** [ˈθɹu]
   - write out actually pronounced sounds
     ne**x**t [ˈnɛkst], e**x**am [əg.ˈzæm]
4. a dot (.) indicates **syllable divisions**: pronunciation [prə.ˌnʌn.si.ˈej.ʃən]
5. **primary** and **secondary** stress are indicated with ˈ and ˌ **before** the stressed syllable

# Consonant and Vowels

- the segments can be roughly divided into **three** groups - **vowels**, **consonants** and **glides**
- the various parts of the vocal tract (vocal folds, tongue, lips, etc.) can create different ‘filters’ that modulate how air passes through
  - **vowels**: little to no obstruction of the airstream, always **voiced**
  - **consonants**: some degree of obstruction of the airstream, **voicing** depends
  - **glides**: these sounds are between vowels and consonants

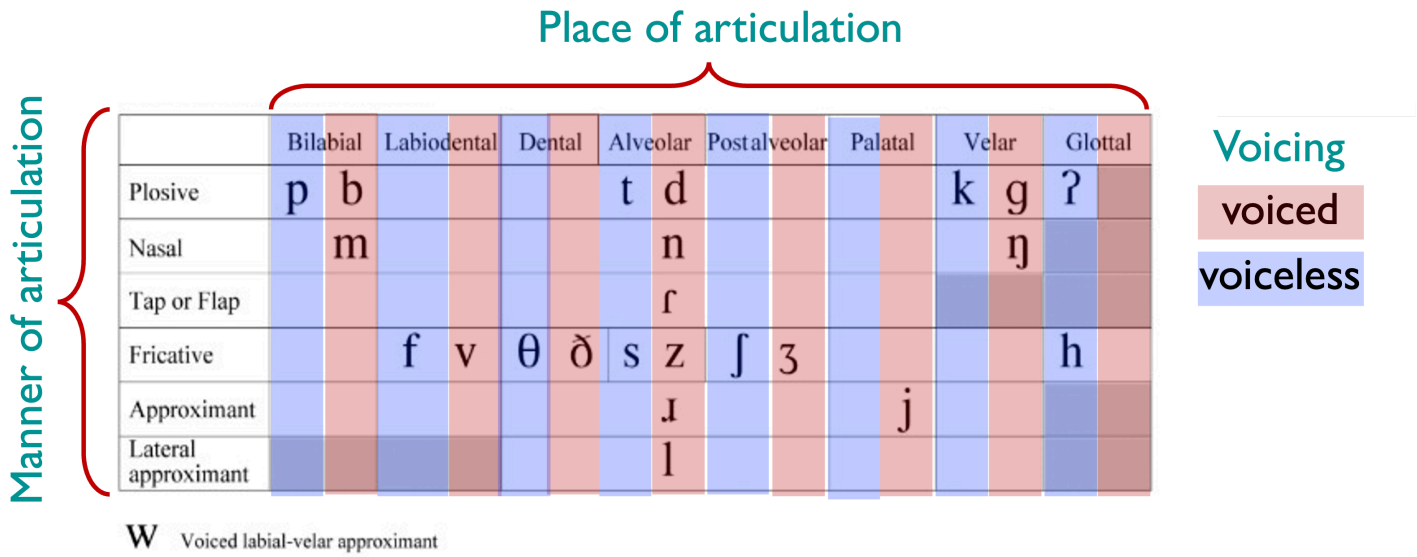


*a consonant's airflow must have obstruction*

# Consonants

- all consonants are produced by restricting or obstructing the airflow in some way – whether partially or totally
- but the nature of the obstruction **varies** from consonant to consonant

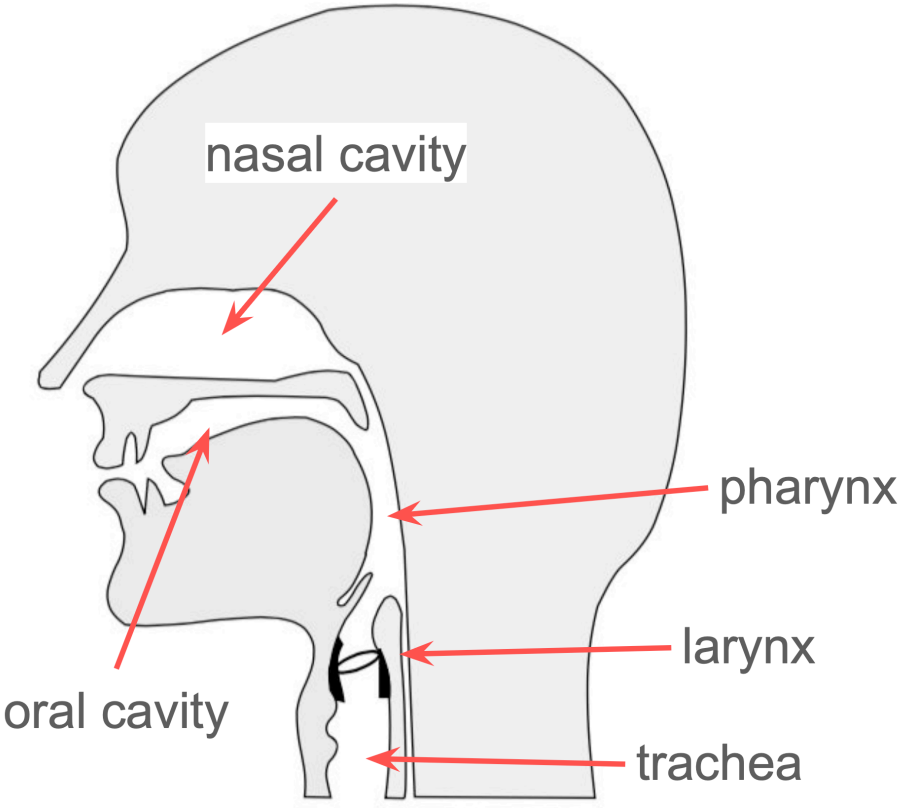
- consonants can be characterized with three properties – which all serve to describe that obstruction
  1. **voicing**: presence or absence of vocal fold vibration
  2. **place of articulation**: where the obstruction of airflow takes place
  3. **manner of articulation**: how the airflow is modulated



# Anatomy of the Vocal Tract

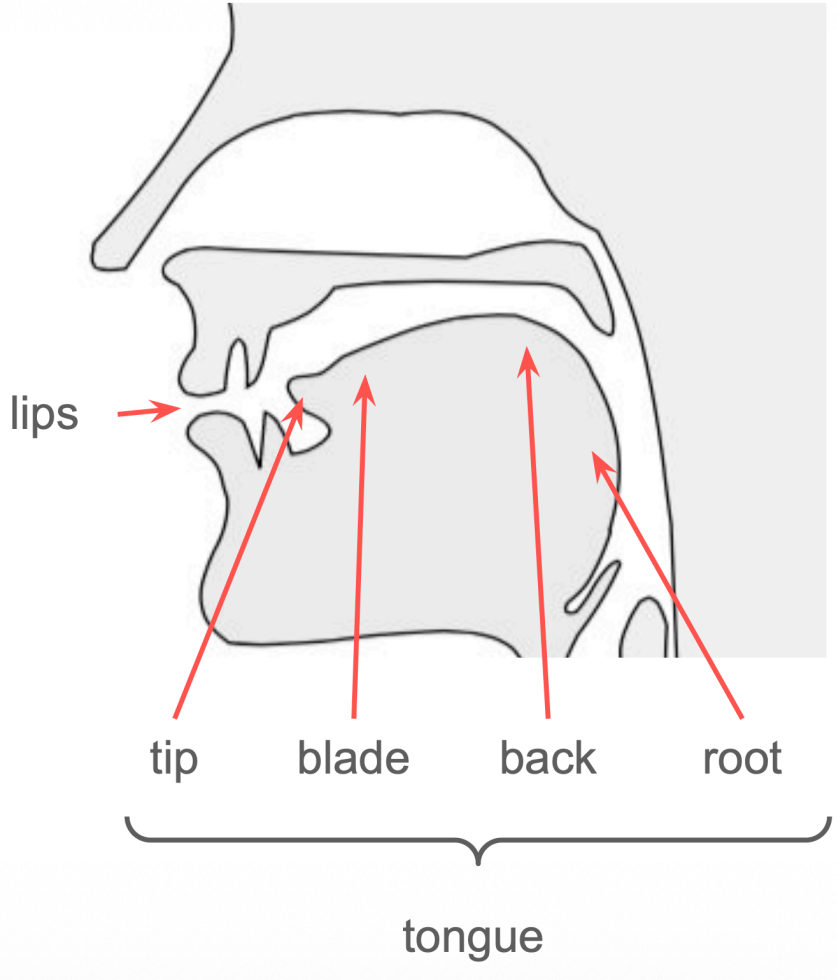
as we exhale:

1. air flows up from our lungs via the **trachea**
2. through the **larynx** (source of **voicing**)
3. through the **pharynx** (open space behind tongue)
4. through the **oral cavity** (mouth) and/or **nasal cavity** (nasal sinuses)
5. and out of body



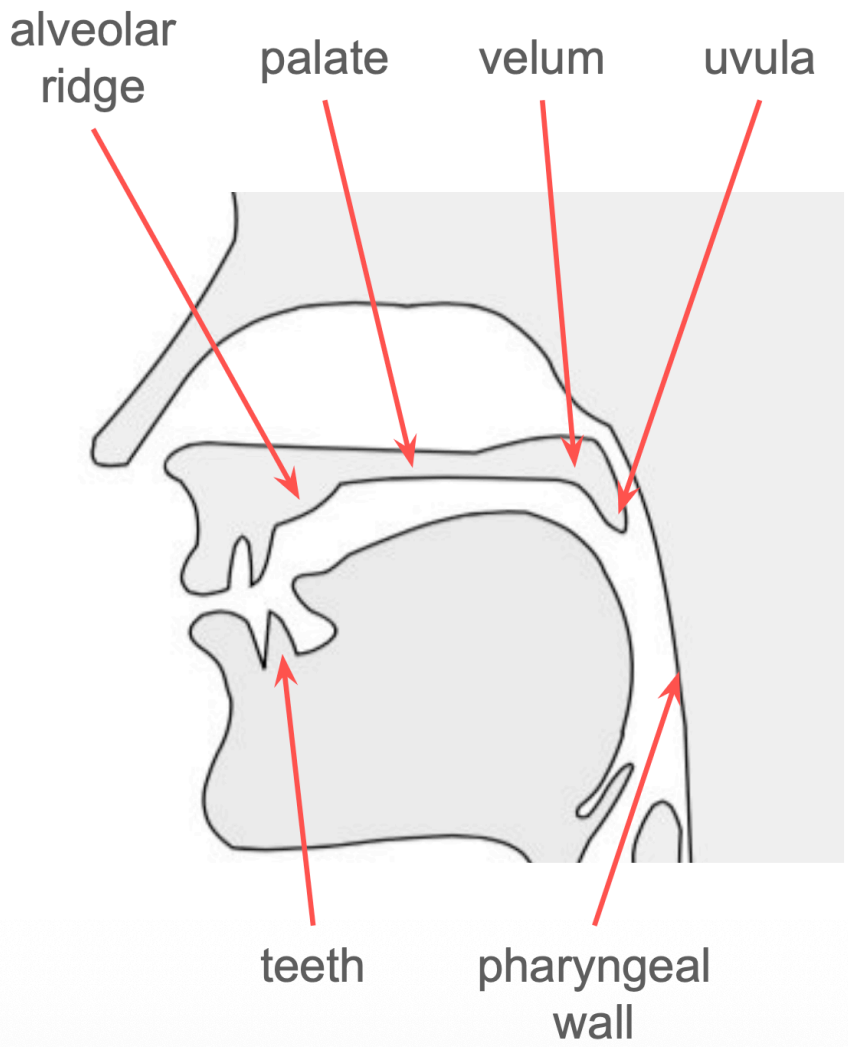
# Anatomy of the Vocal Tract

- to make a consonant, airflow must be **obstructed** (somewhere and somehow)
- this is realised by moving **active articulator** toward a **passive articulator** to restrict airflow
- active articulators in clude **lips** and **tongue**
- **tongue** is further divded into four areas
  - tongue tip
  - tongue blade
  - tongue back
  - tongue root



# Anatomy of the Vocal Tract

- and the rest are the **passive articulators**
  - teeth
  - alveolar ridge
  - hard palate
  - velum (soft palate)
  - uvula
  - pharyngeal wall



# Voicing

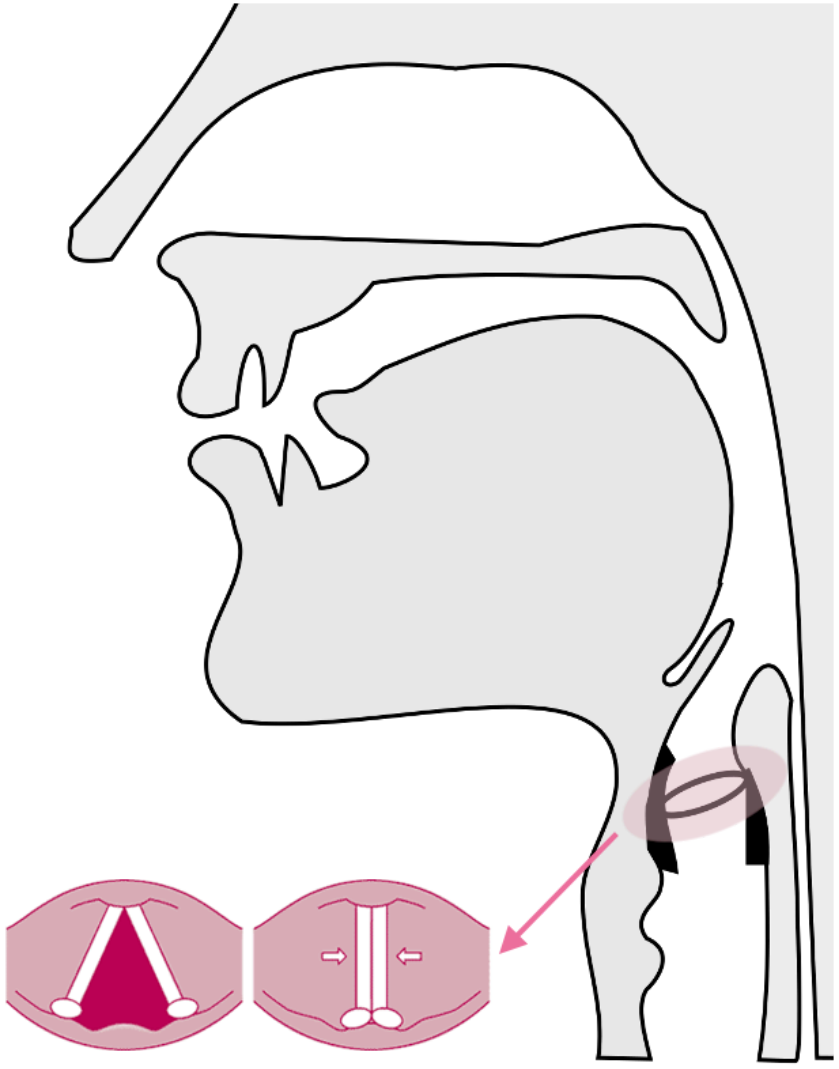
**voiceless**: vocal folds pulled apart and are not vibrating; air goes right through

- **[p]** in ***p****ark*
- **[t]** in ***t****art*
- **[k]** in ***c****ard*

**voiced**: vocal folds brought close together and are vibrating

- **[b]** in ***b****ark*
- **[d]** in ***d****ark*
- **[g]** in ***g****uard*

try pronouncing the sequences fffffvvvvvv and ssssssszzzzzz



# Voicing

(whether vocal cords vibrate or not)



# Place of Articulation

## (where the obstruction is in the vocal tract)

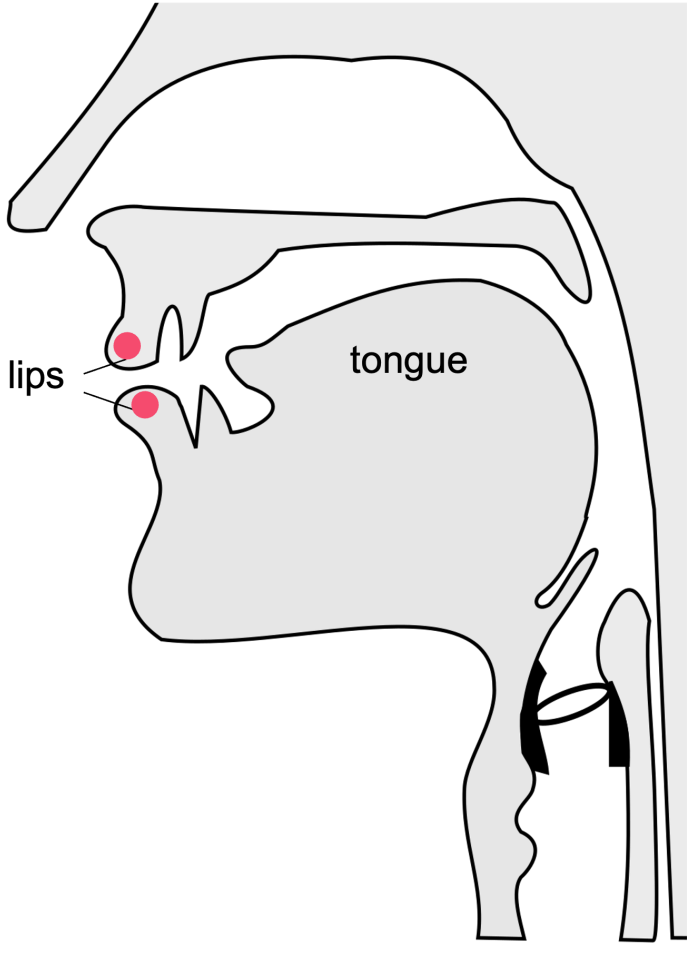
1. **bilabial**: obstruction occurs at the **lips**

| | Bilabial |
|---|---|
| Plosive | p b |
| Nasal | m |
| Trill | ʙ |
| Tap or Flap | |
| Fricative | ɸ β |
| Lateral fricative | |
| Approximant | |
| Lateral approximant | |

< **m**an > **[ m ]**

< **b**an > **[ b ]**

< **p**an > **[ p ]**



# Place of Articulation

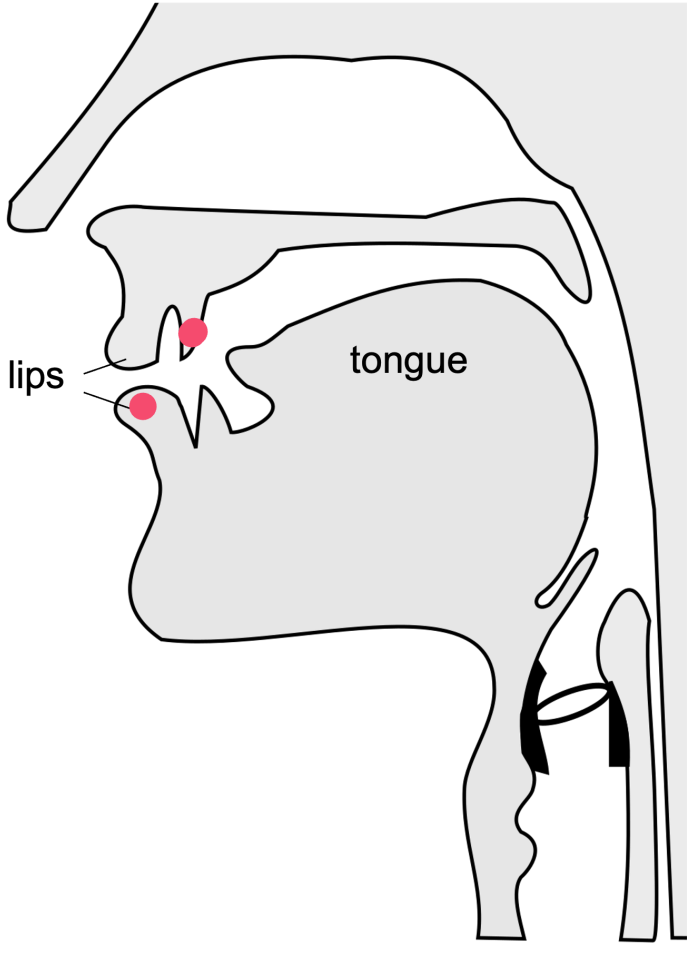
## (where the obstruction is in the vocal tract)

2. **labiodental**: obstruction occurs with the **lower lip** and the **upper teeth**

| | Labiodental |
|---|---|
| Plosive | |
| Nasal | ɱ |
| Trill | |
| Tap or Flap | ⱱ |
| Fricative | f v |
| Lateral fricative | |
| Approximant | ʋ |
| Lateral approximant | |

< fan > [ f ]

< van > [ v ]



# Place of Articulation

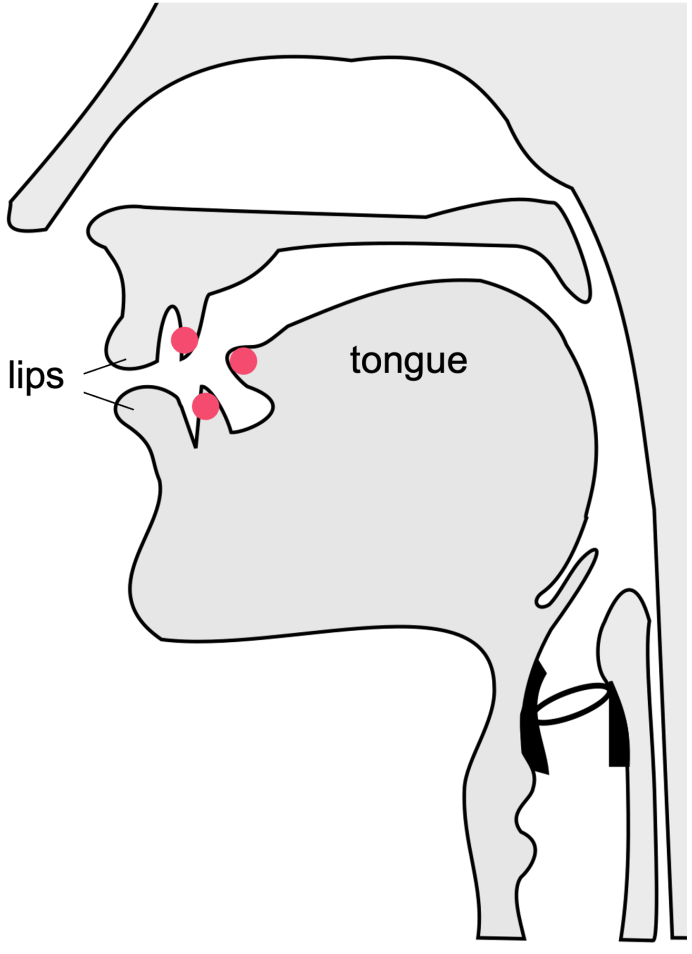
## (where the obstruction is in the vocal tract)

3. **dental/interdental**: obstruction occurs when the **tongue tip** is between the **teeth**

| | Dental |
|---|---|
| Plosive | |
| Nasal | |
| Trill | |
| Tap or Flap | |
| Fricative | θ ð |
| Lateral fricative | |
| Approximant | |
| Lateral approximant | |

< **th**is > **[ ð ]**

< **th**ick > **[ θ ]**



# Place of Articulation

## (where the obstruction is in the vocal tract)

4. **alveolar**: obstruction occurs when the **tongue tip** raises toward the **alveolar ridge** behind the teeth

| | Alveolar |
|---|---|
| Plosive | t d |
| Nasal | n |
| Trill | r |
| Tap or Flap | ɾ |
| Fricative | s z |
| Lateral fricative | ɬ ɮ |
| Approximant | ɹ |
| Lateral approximant | l |

< **d**ate > [ d ]

< **t**ight > [ t ]

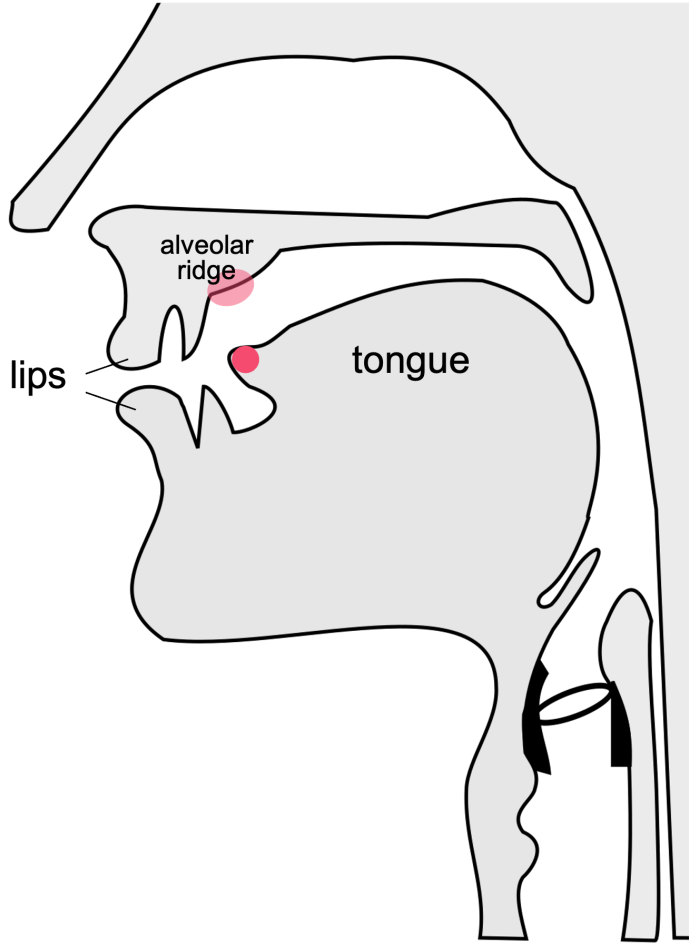
< **n**ate > [ n ]

< bu**tt**er > [ ɾ ]

< **s**ate > [ s ]

< **l**ate > [ l ]

< **r**ate > [ ɹ ]



# Place of Articulation

## (where the obstruction is in the vocal tract)

5. **postalveolar**: obstruction occurs when the **tongue tip** raises between the **alveolar ridge** and the **palate**

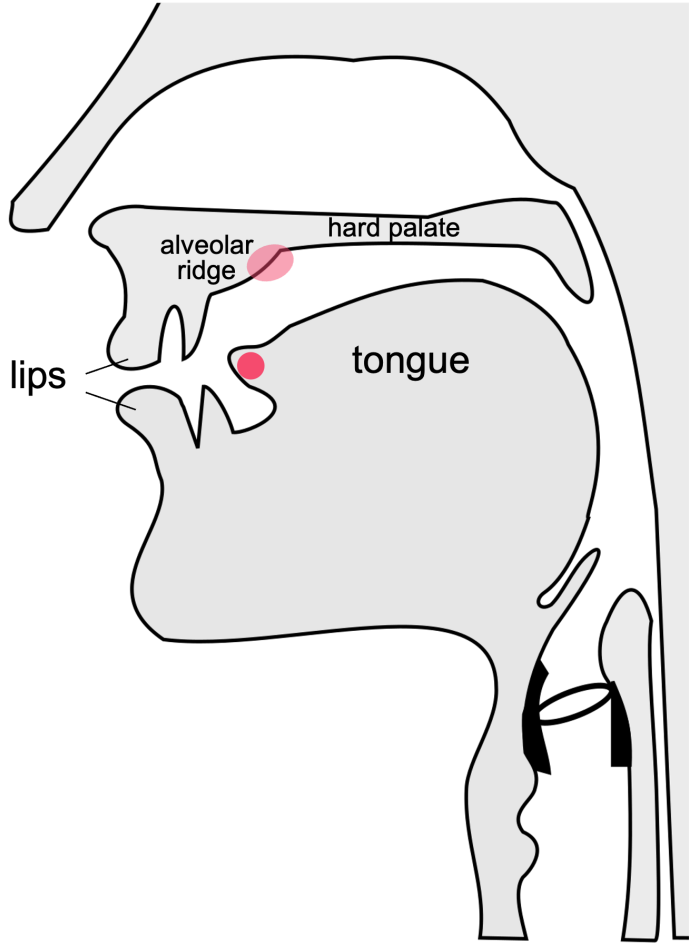
| | Postalveolar |
|---|---|
| Plosive | |
| Nasal | |
| Trill | |
| Tap or Flap | |
| Fricative | ʃ ʒ |
| Lateral fricative | |
| Approximant | |
| Lateral approximant | |

< rouge > [ ʒ ]

< crush > [ ʃ ]

< cheat > [ tʃ ]

< age > [ dʒ ]




tʃ: voiceless palato-alveolar **affricate**

dʒ: voiced palato-alveolar **affricate**

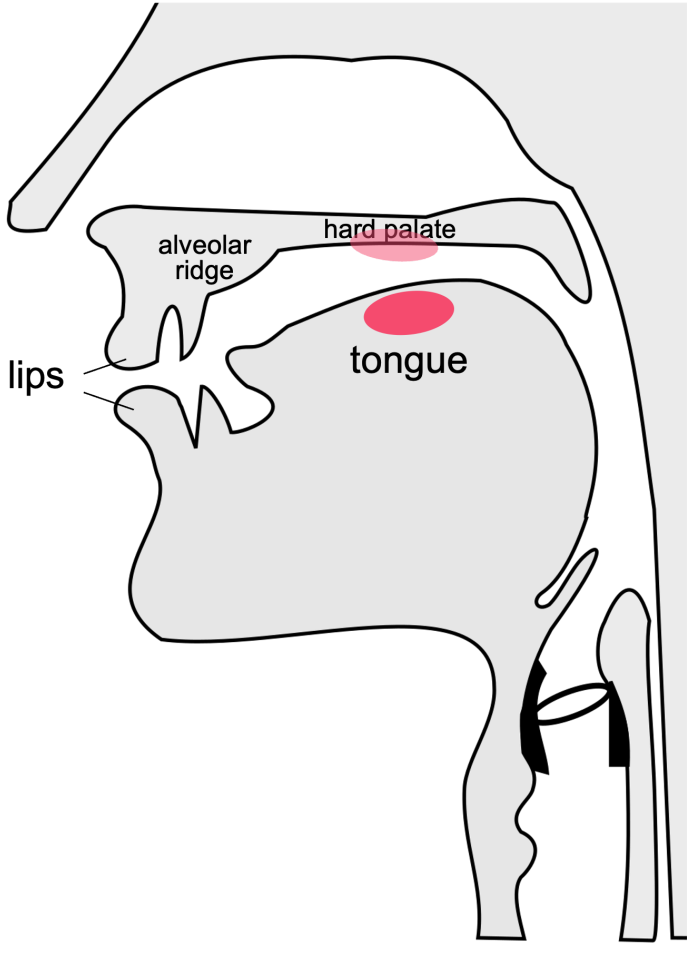
# Place of Articulation

## (where the obstruction is in the vocal tract)

6. **palatal**: obstruction occurs when the **tongue body** raises towards the **palate**

| | Palatal |
|---|---|
| Plosive | c ɟ |
| Nasal | ɲ |
| Trill | |
| Tap or Flap | |
| Fricative | ç ʝ |
| Lateral fricative | |
| Approximant | j |
| Lateral approximant | ʎ |

< yes > [ j ]



# Place of Articulation

## (where the obstruction is in the vocal tract)

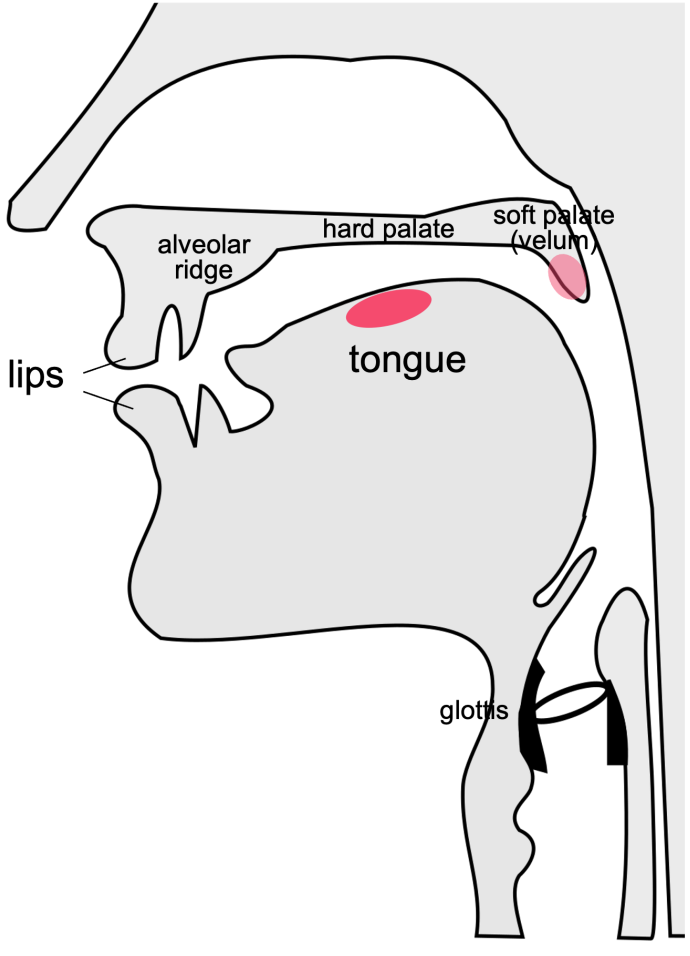
7. **velar**: obstruction occurs when the **tongue body** raises towards the **velum**

| | Velar |
|---|---|
| Plosive | k g |
| Nasal | ŋ |
| Trill | |
| Tap or Flap | |
| Fricative | x ɣ |
| Lateral fricative | |
| Approximant | ɯ̟ |
| Lateral approximant | ʟ |

< coat > [ k ]

< goat > [ g ]

< sing > [ ŋ ]



# Place of Articulation

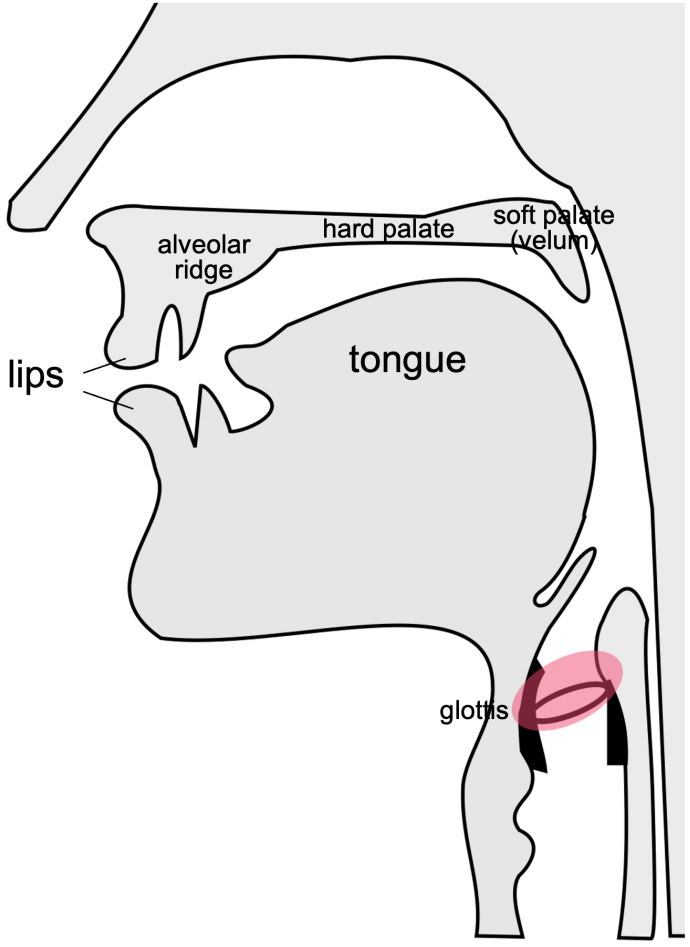
## (where the obstruction is in the vocal tract)

8. **glottal**: obstruction occurs at the **glottis**

| | Glottal |
|---|---|
| Plosive | ʔ |
| Nasal | |
| Trill | |
| Tap or Flap | |
| Fricative | h ɦ |
| Lateral fricative | |
| Approximant | |
| Lateral approximant | |

< a**h**a > **[ h ]**

< **u**h-**o**h > **[ ? ]**



# Place of Articulation

## (where the obstruction is in the vocal tract)

the IPA chart is organized so that the place of articulation progresses from the front of the **vocal tract** to the back

**front** **back**

| | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Plosive | p b | | | t d | | ʈ ɖ | c ɟ | k g | q ɢ | | ʔ |
| Nasal | m | ɱ | | n | | ɳ | ɲ | ŋ | ɴ | | |
| Trill | ʙ | | | r | | | | | ʀ | | |
| Tap or Flap | | ⱱ | | ɾ | | ɽ | | | | | |
| Fricative | ɸ β | f v | θ ð | s z | ʃ ʒ | ʂ ʐ | ç ʝ | x ɣ | χ ʁ | ħ ʕ | h ɦ |
| Lateral fricative | | | | ɬ ɮ | | | | | | | |
| Approximant | | ʋ | | ɹ | | ɻ | j | ɰ | | | |
| Lateral approximant | | | | l | | ɭ | ʎ | ʟ | | | |

- English does NOT exhibit above places of articulation for consonants (marked red)

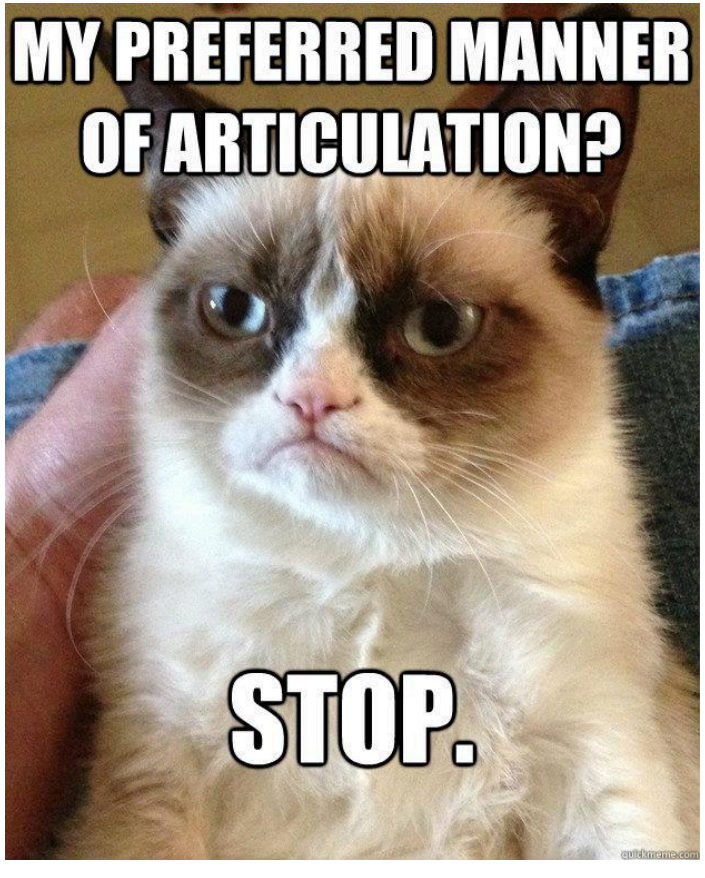
# Manner of Articulation

## (what type of obstruction of airflow occurs)

**complete blockage of airflow**

- stops/plosives **[p/b, t/d, k/g]**
  - < park >, [ **p** ]
  - < bark >, [ **b** ]
  - < tart >, [ **t** ]
  - < dark >, [ **d** ]
  - < card >, [ **k** ]
  - < guard >, [ **g** ]

- trills, taps/flaps **[r, ɾ]**
  - obstruction doesn't last long enough to cause a **release of pressure** at the end
  - < butter >, [ **ɾ** ]



# Manner of Articulation

## (what type of obstruction of airflow occurs)

**partial blockage of airflow**

- fricatives [ʃ, ʒ]
  - < tension >, [ ʃ ]
  - < lesion >, [ ʒ ]
- approximants [ɹ]
  - less obstruction than **fricatives**
  - < red >, [ ɹ ]
- laterals [l]
  - obstruction in the center, but air flows past the tongue on the side(s)
  - < lesion >, [ l ]



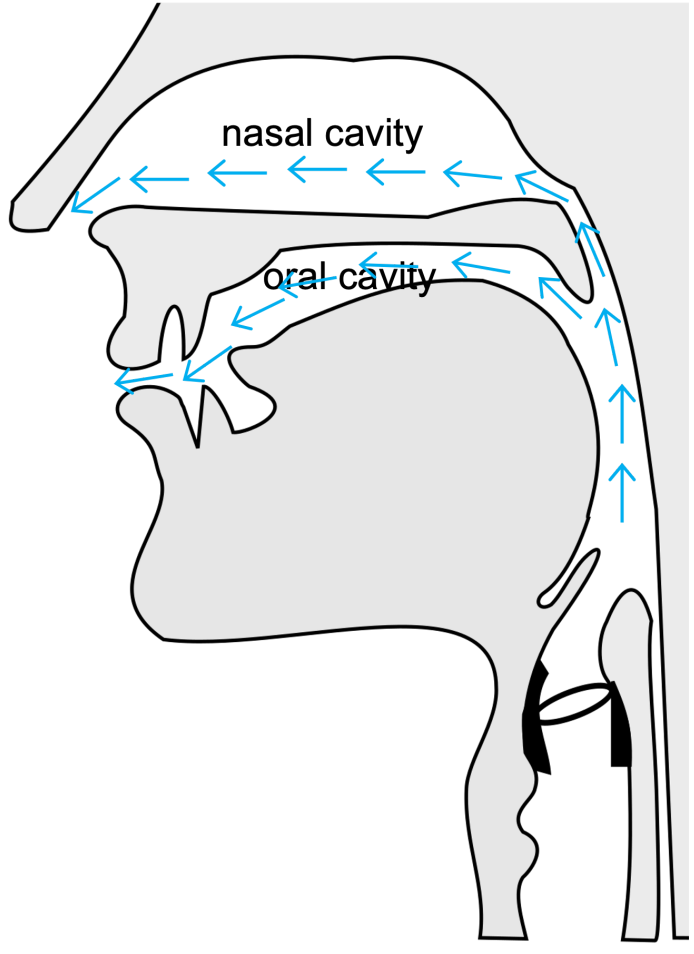
# Manner of Articulation

## (what type of obstruction of airflow occurs)

airflow blocked in **one part** of the vocal tract but flow through **a different part**

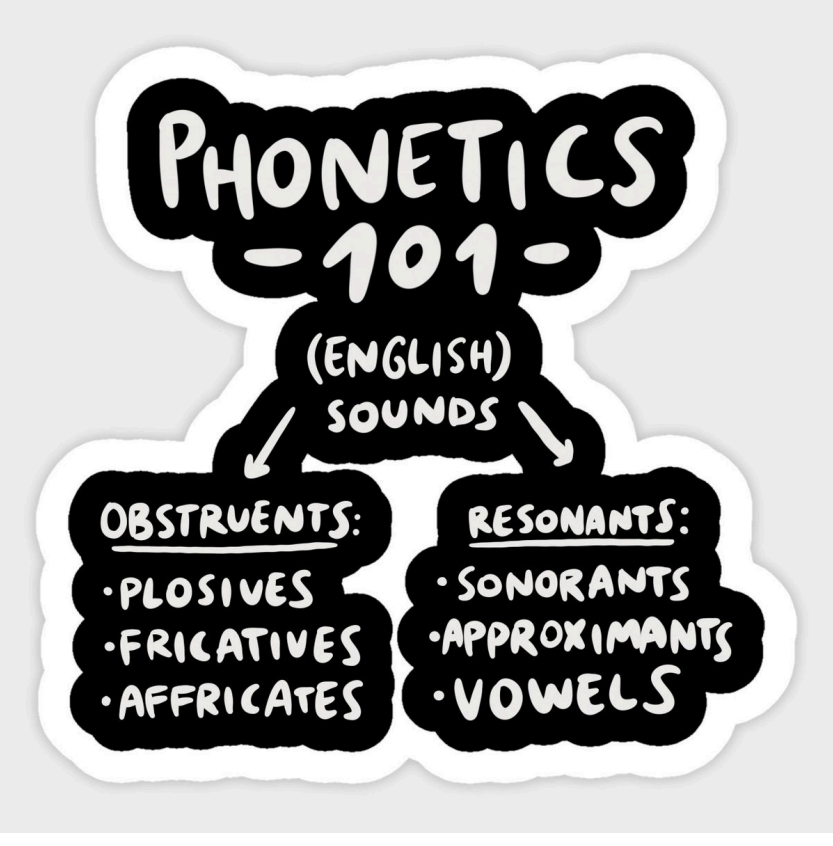
- affricates **[t͡ʃ, d͡ʒ]** ???[d͡ʃ]
  - a combination of stops and fricatives
  - < **ch**oke >, [ **t͡ʃ** ]
  - < **j**oke >, [ **d͡ʒ** ]

- nasals **[m, n, ŋ]**
  - complete obstruction through the oral cavity but air is **redirected** to flow through the nasal cavity with **lowered velum**
  - < **m**ail >, [ **m** ]
  - < **n**ail >, [ **n** ]
  - < si**ng** >, [ **ŋ** ]



# Manner of Articulation

## (what type of obstruction of airflow occurs)




further grouping into **sound class**

- rhotics: ***r***-like sounds **[ɹ, ɻ, r, ʀ, ɾ, ɽ, ʁ...]**
- liquids: rhotics and laterals
- obstruents
  - consonants with **more** obstruction
  - stops, fricatives, affricates
- sonorants
  - consonants with **less** obstruction
  - nasals, liquids, approximants

# One More Property: Aspiration

compare **< space >** and **< pace >**

compare **< scare >** and **< care >**

- any difference in the **< p >** s and **< c >** s?
- put a piece of paper in front of you and pronounce the sound

**aspiration**: a puff of air

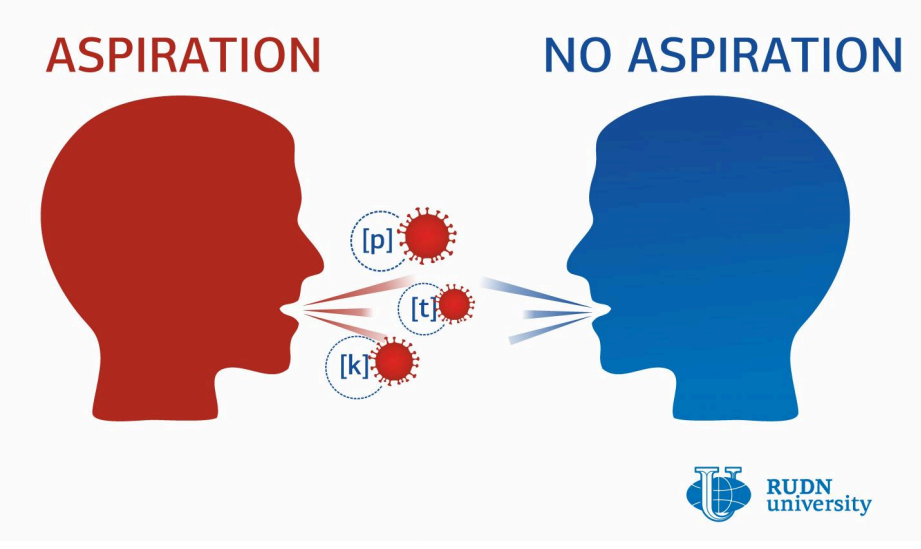
- **[$p^h$]** vs **[p]**
- **[$k^h$]** vs **[k]**

NEWS RELEASE 30-AUG-2020

Aspirated consonants may promote the spread of COVID-19, RUDN University linguist says

Peer-Reviewed Publication

RUDN UNIVERSITY



# Feel Consonants

**stops/plosives** - **fricatives**: try to pronounce a **long** sequence of a consonant

- if it can be pronounced with non-stop, the sound is a **fricative**
- if it cannot be pronounced with non-stop, the sound is a **stop**



**oral** - **nasal**: hold **nose** and produce the sound

- if it becomes hard to produce, the sound is a **nasal**
- if it still sounds similar, the sound is a **oral**



try these tests on the first sound in < fan >, < pan > and < man >

# Feel Consonants: Notes about Affricates

- say the words ***t**ie*, ***sh**y*, ***ch**ai* and do the **stop - fricative** test
  - with which words does your tongue **touch** the upper part of your mouth?
  - which words produce a hissing sound?

- **affricates** are a sort of combination between stops and fricatives
  - a complete closure is made, building up pressure
  - when it is released, the articulators are kept close together enough to produce the hissing sound characteristic of fricatives

- affricates are considered **one sound**, but with two phases, and they are written with two IPA symbols joined with a **tie bar**
  - *c**h**eck* [t͡ʃɛk]



+



# Exercise: Feel Consonants

**differentiate** the following consonants considering properties we have mentioned above

**fricative - stop**
**voiced - voiceless**

**oral - nasal**
**aspirated - unaspirated**

| sound | transcription | fricative/stop? | oral/nasal? | voiced/voiceless? | aspirated/unaspirated? |
|---|---|---|---|---|---|
| 1st in **< phone >** | | | | | |
| 1st in **< think >** | | | | | |
| 1st in **< tiger >** | | | | | |
| 1st in **< milk >** | | | | | |
| 1st in **< cat >** | | | | | |
| 3rd in **< vision >** | | | | | |

# Exercise: Feel Consonants

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| sound | transcription | fricative/stop? | oral/nasal? | voiced/voiceless? | aspirated/unaspirated? |
|---|---|---|---|---|---|
| 1st in **< phone >** | **[f]** | | | | |
| 1st in **< think >** | **[θ]** | | | | |
| 1st in **< tiger >** | **[$t^h$]** | | | | |
| 1st in **< milk >** | **[m]** | | | | |
| 1st in **< cat >** | **[$k^h$]** | | | | |
| 3rd in **< vision >** | **[ʒ]** | | | | |

# Exercise: Feel Consonants

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|---|---|---|---|---|---|
| 1st in **< phone >** | **[f]** | **fricative** | | | |
| 1st in **< think >** | **[θ]** | **fricative** | | | |
| 1st in **< tiger >** | **[$t^h$]** | **stop** | | | |
| 1st in **< milk >** | **[m]** | **stop** | | | |
| 1st in **< cat >** | **[$k^h$]** | **stop** | | | |
| 3rd in **< vision >** | **[ʒ]** | **fricative** | | | |

# Exercise: Feel Consonants

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| sound | transcription | fricative/stop? | oral/nasal? | voiced/voiceless? | aspirated/unaspirated? |
|---|---|---|---|---|---|
| 1st in **< phone >** | **[f]** | fricative | **oral** | | |
| 1st in **< think >** | **[θ]** | fricative | **oral** | | |
| 1st in **< tiger >** | **[$t^h$]** | stop | **oral** | | |
| 1st in **< milk >** | **[m]** | stop | **nasal** | | |
| 1st in **< cat >** | **[$k^h$]** | stop | **oral** | | |
| 3rd in **< vision >** | **[ʒ]** | fricative | **oral** | | |

# Exercise: Feel Consonants

**differentiate** the following consonants considering properties we have mentioned above

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**voiced - voiceless**

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| sound | transcription | fricative/stop? | oral/nasal? | voiced/voiceless? | aspirated/unaspirated? |
|---|---|---|---|---|---|
| 1st in **< phone >** | **[f]** | fricative | oral | **voiceless** | |
| 1st in **< think >** | **[θ]** | fricative | oral | **voiceless** | |
| 1st in **< tiger >** | **[$t^h$]** | stop | oral | **voiceless** | |
| 1st in **< milk >** | **[m]** | stop | nasal | **voiced** | |
| 1st in **< cat >** | **[$k^h$]** | stop | oral | **voiceless** | |
| 3rd in **< vision >** | **[ʒ]** | fricative | oral | **voiced** | |

# Exercise: Feel Consonants

**differentiate** the following consonants considering properties we have mentioned above

**fricative - stop**
**voiced - voiceless**

**oral - nasal**
**aspirated - unaspirated**

| sound | transcription | fricative/stop? | oral/nasal? | voiced/voiceless? | aspirated/unaspirated? |
|---|---|---|---|---|---|
| 1st in **< phone >** | **[f]** | fricative | oral | voiceless | **unaspirated** |
| 1st in **< think >** | **[θ]** | fricative | oral | voiceless | **unaspirated** |
| 1st in **< tiger >** | **[$t^h$]** | stop | oral | voiceless | **aspirated** |
| 1st in **< milk >** | **[m]** | stop | nasal | voiced | **unaspirated** |
| 1st in **< cat >** | **[$k^h$]** | stop | oral | voiceless | **aspirated** |
| 3rd in **< vision >** | **[ʒ]** | fricative | oral | voiced | **unaspirated** |

**Assignment VI** is due this Friday (**Nov 13th**)

**Assignment VII** will be published this Wednesday (**Nov 11**) and is due next Friday (**Nov 20th**) **continue reading**: Phonetics in O'Grady et al.'s *Contemprary Linguistics: An Introduction*

This is Schwa.

Schwa is not stressed.

Schwa is cool.

Be like Schwa.



Slides created via the R package **xaringan**.